Assignment No. 9

Title: Implementation of Moving Average (Weights – Simple Moving Average)

Objective:

To implement the Moving Average technique using Python for time series analysis and forecasting. The objective is to smooth the fluctuations in the dataset and identify trends in the data using the Simple Moving Average method.

Problem Statement:

Implement the Moving Average Method using Python on a time series dataset. Load the dataset into a pandas DataFrame and apply the Simple Moving Average technique to smooth the data and analyze patterns over time. Visualize the results using graphs to compare the original data and the moving average values.

Theory:

Moving Average Method

The Moving Average method is a statistical technique used in time series analysis to smooth out short-term fluctuations and highlight longer-term trends or patterns in the data.

In this method, the average of a fixed number of consecutive observations is calculated. As new data becomes available, the oldest observation is removed and the average is recalculated. This process results in a moving window that continuously updates the average values.

Moving Average methods are commonly used in financial analysis, stock market prediction, weather forecasting, and economic data analysis.

Simple Moving Average (SMA)

The Simple Moving Average is the most basic form of moving average. It calculates the arithmetic mean of a specified number of past observations.

Mathematical Formula

$$SMA_t = \frac{Y_t + Y_{t-1} + Y_{t-2} + ... + Y_{t-n+1}}{n}$$

Where:

- SMA_t = Simple Moving Average at time t
- Y_t = Current value
- n = Number of time periods (window size)

The SMA helps remove noise from the data and provides a clearer view of the underlying trend.

Applications of Moving Average

1. Stock market trend analysis
2. Sales forecasting
3. Weather data analysis
4. Economic forecasting
5. Signal processing

Implementation using Python

Step 1: Import Required Libraries

Python libraries such as Pandas and Matplotlib are used for data analysis and visualization.

```
import pandas as pd
import matplotlib.pyplot as plt
```

- **Pandas** is used for handling and analyzing data.
- **Matplotlib** is used for plotting graphs.

Step 2: Load the Dataset

The dataset is loaded using the pandas **read_csv()** function.

```
df = pd.read_csv("data.csv")
```

This loads the dataset into a pandas DataFrame for further processing.

Step 3: Convert Date Column into Datetime Format

```
df['Date'] = pd.to_datetime(df['Date'])
```

This converts the Date column into datetime format, which is necessary for time series analysis.

Step 4: Set Date Column as Index

```
df.set_index('Date', inplace=True)
```

The Date column is set as the index to organize the dataset based on time.

Step 5: Calculate Simple Moving Average

```
df['SMA_5'] = df['Close'].rolling(window=5).mean()
```

This calculates the **5-period Simple Moving Average**.

- `rolling(window=5)` creates a moving window of size 5.

- `mean()` calculates the average of the values within that window.

Step 6: Calculate Another Moving Average

```
df['SMA_20'] = df['Close'].rolling(window=20).mean()
```

This calculates a **20-period moving average**, which helps identify longer-term trends.

Step 7: Plot the Results

```
plt.figure(figsize=(12,6))
plt.plot(df['Close'], label="Original Data")
plt.plot(df['SMA_5'], label="5-Day Moving Average")
plt.plot(df['SMA_20'], label="20-Day Moving Average")
plt.legend()
plt.title("Simple Moving Average Method")
plt.xlabel("Date")
plt.ylabel("Value")
plt.show()
```

This graph compares the **original dataset** with the **moving average values**, making it easier to identify trends.

Advantages of Simple Moving Average

1. Easy to understand and implement.
2. Helps smooth noisy time series data.
3. Useful for identifying trends and patterns.
4. Widely used in financial and stock market analysis.

Limitations of Simple Moving Average

1. Gives equal weight to all observations.
2. May lag behind actual data trends.
3. Not effective for datasets with sudden fluctuations.

Conclusion

Thus, the Moving Average Method using Simple Moving Average was successfully implemented using Python. The technique helped smooth short-term fluctuations in the dataset and identify the underlying trend in the data. Moving Average methods are widely used in time series analysis and forecasting applications.